

Dushan Kavishka

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Education

Master of Science in Civil Engineering

May 2027

University of North Florida, Jacksonville, FL

- **Current GPA:** 4.00 / 4.00.
- **Relevant Coursework:** Computational Fluid Mechanics, Floating Structures, Coastal Structures, Numerical Modeling, Aquatic Chemical Processes.

Bachelor of Science in Civil Engineering (Honors) - Washington Accord Accredited

Jul 2024

University of Moratuwa, Colombo, Sri Lanka

- **Overall GPA:** 3.45 / 4.00 (Second Class Upper Division).
- **Final Semester GPA:** 3.97 / 4.00 (Dean's List).
- **Relevant Coursework:** Fluid Mechanics, Hydraulic Engineering, Hydraulic Design, Coastal and Port Engineering, Environmental Engineering, Surveying, Bridge Engineering, Geotechnical Engineering, Construction Planning and Cost Estimation, Irrigation Engineering, Applied Statistics.

Technical Skills

Engineering Software: AutoCAD, Civil 3D, Revit, ArcGIS Pro, HEC-HMS, Delft3D, MIKE, ANSYS Fluent

Programming and Analysis: MATLAB, Python, Fortran

Project Tools: Microsoft Office Package, Microsoft Project

Work Experience

Graduate Research Assistant

Aug 2025 – Present

University of North Florida, Jacksonville, FL

- Develop and calibrate Delft3D numerical model to simulate hydro- and morphodynamic processes of a Florida barrier island under hurricane wave conditions.
- Utilize MATLAB and Python for model setup, data processing, analysis, and visualization of simulation results.

Research Engineer (Water Resources and Coastal)

Oct 2024 – Aug 2025

Lanka Hydraulic Institute Ltd (LHI), Colombo, Sri Lanka

- Contributed to three major fishery harbor development projects in Oman (Lakbi, Hadbin, Haramel), performing 2D flume and 3D basin physical modeling tests.
- Conducted numerical simulations using MIKE to evaluate wave propagation and harbor calmness.
- Assisted in geotechnical investigations and hydrographic surveys for port and coastal infrastructure projects, including ASYAD Drydock at Duqm Port.
- Prepared detailed technical reports and presented findings during client coordination meetings.

Civil Engineering Intern

Jan 2023 – May 2023

Maga Engineering (Pvt) Ltd, Colombo, Sri Lanka

- Developed strong technical skills in project planning, site inspection, and construction methodologies while developing proficiency in AutoCAD, Civil3D, Revit, Microsoft Office, and Microsoft Project.
- Effectively supervised workers to ensure on-site safety, productivity, and compliance with client requirements.

Projects and Publications

- **Final Year Undergraduate Project** - planning and design of a resort in Sri Lanka, including an Environmental Impact Assessment (EIA), and design of a steel factory building using SAP2000.
- **Undergraduate Research Project** - Numerical investigation of wind loading on gable-roof, low-rise buildings using ANSYS Fluent for CFD simulations. Won the best scientific paper award at the special session on protective structure designs, ICSBE 2024 Conference (DOI: 10.1007/978-981-96-8301-7.21).
- **Conferences** - Ongoing Master's research has been presented at the 2026 UF Water Institute Symposium and is scheduled for presentation at ICCE 2026 in May.